5.6.3 Electrical Energy

After completing this chapter you should be able to apply models, theories and laws to situations involving energy, force and motion.

For Questions 1 to 10 choose the best answer (A), (B), (C) or (D).

1. Which statement is true?

(A) A current will flow in a circuit if it has a break in it.
(B) Current flows in a circuit only when the switch is open.
(C) If a circuit does not have a switch it will not work.
(D) Ammeters measure the current flowing in a circuit.

2. What do we mean by components connected in series?

(A) The components are connected one after the other.
(B) Each component is connected to all the other components.
(C) The components are connected in the correct order.
(D) All components are connected to the battery.

3. What is the unit we use to measure electrical current?

(A) A.
(B) Ampere.
(C) Ammeter.
(D) Ohm.

4. If a circuit is closed, then:

(A) The switch will be off.
(B) Any light globes in it will be glowing.
(C) An ammeter in it will read zero current.
(D) One of the leads could be disconnected.

5. Which statement about correctly wired circuits containing several globes connected in series, is correct?

(A) The more globes in the circuit, the brighter they will glow.
(B) Globes furthest from the power supply will glow less brightly.
(C) All globes in the circuit will glow equally bright.
(D) The fewer globes there are in the circuit, the dimmer they will glow.

6. Which statement is correct?

(A) Electrons are held more tightly to their atoms in conductors than in resistors.
(B) Electrons are held more tightly to their atoms in conductors than in insulators.
(C) Electrons are held more tightly to their atoms in resistors than in insulators.
(D) Electrons are held more tightly to their atoms in insulators than in conductors.

7. Why do we use resistors in electrical circuits?

(A) To reduce the amount of electricity flowing through the circuit.
(B) To control the amount of electricity flowing through the circuit.
(C) To reduce the amount of energy changing into other forms in the circuit.
(D) To change the electricity flowing through the circuit into other forms of energy.

8. Why is air not used inside light globes?

(A) The metal in the filament would rust.
(B) The metal in the filament would burn.
(C) The metal in the filament would melt.
(D) Air does not glow as much as the argon/nitrogen mixture used.

9. What is another name for voltage?

(A) Electrical potential energy difference.
(B) Electrostatic power.
(C) Electrical power.
(D) Electrical potential energy.

10. Which choice best describes what happens if the voltage applied to a circuit is increased?

(A) The resistance of the circuit increases.
(B) The resistance of the circuit decreases.
(C) The current flowing in the circuit increases.
(D) The current flowing in the circuit decreases.

11. Complete the following table:

Amperes	Milliamperes	Microamperes
1		
	500	
		1 500 000
2.5		
	6500	
		15 000

12. Explain the difference between an open and a closed circuit.

.....

.....

.....

.....

.....

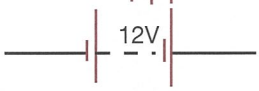
Science Press

13. Identify each of the following electrical circuit symbols.

(a) 

(b) 

(c) 

(d) 

(e) 

(f) 

(g) 

(h) 

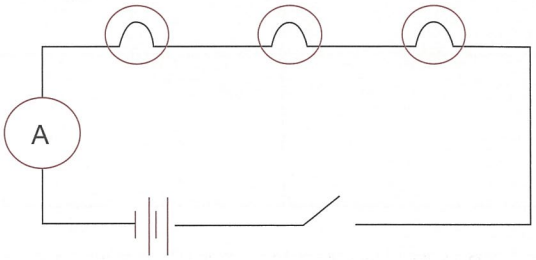
(i) 

(j) 

14. What hidden message is in the statement 'carelessness with electricity probably only happens once in your life'?

.....

15. In your own words, describe the circuit shown.



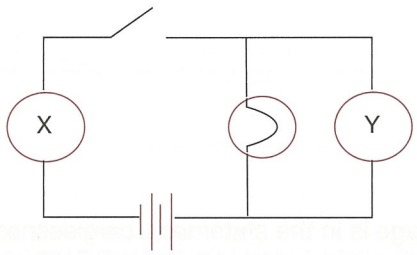
.....

.....

.....

16. A circuit contains two resistors connected in series with each other, a 12 V battery and a closed switch. Draw this circuit in the space provided.

17. The diagram shows a correct circuit. Complete the table to give the required information about meters X and Y.



Meter	Type of meter	How connected into circuit	Function of meter
X			
Y			

18. An electrical circuit is often compared to the system of pipes and pumps that supply water to our homes. Outline how the voltage, resistance, a switch and current are represented in the water system model.

.....

.....

.....

.....

.....

.....

19. Identify one situation in your house where:

(a) A series circuit is used. How do you know this?

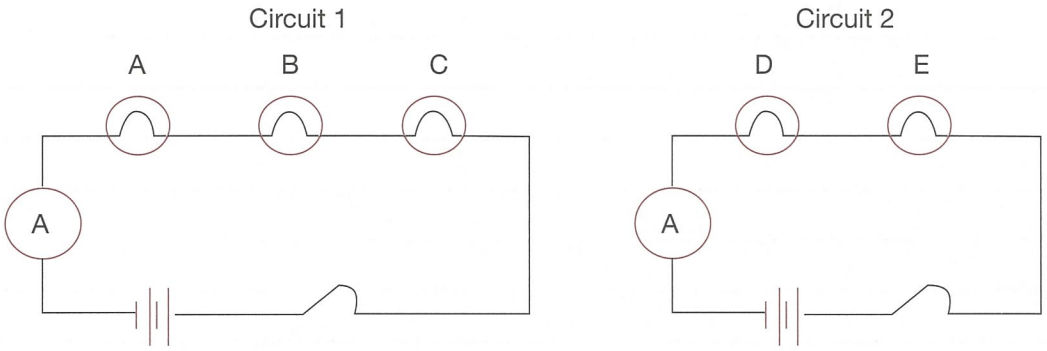
.....

(b) A parallel circuit is used. How do you know this?

.....

.....

20. Look at the two circuits below. Five globes are labelled A, B, C, D and E.



(a) Compare the brightness of each globe with all the other globes.

.....

.....

.....

.....

.....

.....

(b) Explain your answer.

.....

.....

(c) Now suppose globe B is removed from circuit X and replaced with a piece of circuit wiring. Compare the brightness of the globes now.

.....

(d) Explain your answer.

.....

.....

Notes